

- 5 (a) Fig. 5.1 is a photomicrograph of human blood cells that have been placed in a solution of sodium chloride.

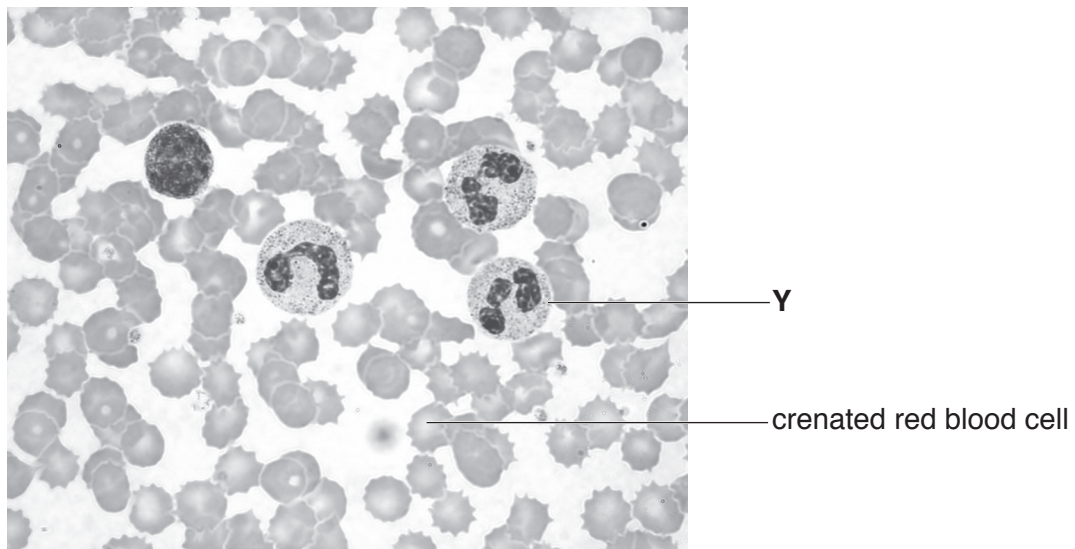


Fig. 5.1

- (i) State the function of the cell labelled Y.

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..... [1]

- (ii) The red blood cell labelled in Fig. 5.1 is described as crenated because it has an abnormal, shrivelled appearance.

Explain how this red blood cell has become crenated.

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..... [2]

The results are shown in Table 5.1.

- (c) Carbonic anhydrase is an enzyme found in red blood cells which has an important role in the transport of respiratory gases.

Explain why a non-competitive inhibitor of carbonic anhydrase will reduce the supply of oxygen to actively respiring tissues.

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..... [2]

[Total: 10]